

A Harmonic View of Random Walks and Electrical Networks

From Dirichlet Problems to Recurrence of $\mathbb{Z}^2$

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Guiding idea

harmonicity $\implies$ uniqueness

$\implies$ random walk / electrical dictionary $\implies$ resistance estimate

$\implies$ Rayleigh monotonicity $\implies \mathbb{Z}^2$ is recurrent.

Main point: the connection is built through harmonic functions.

Discrete harmonicity

Let $G = (V, E)$ be a finite graph.

A function $f : V \rightarrow \mathbb{R}$ is *harmonic at an interior vertex* x if

$$f(x) = \frac{1}{\deg(x)} \sum_{y \sim x} f(y).$$

Equivalently, for simple random walk,

$$f(x) = \mathbb{E}_x[f(X_1)].$$

So harmonicity is a *local averaging law*.

The same averaging law appears on both the probabilistic and electrical sides.

Random walk side: hitting probabilities

Let $A, B \subset V$ be disjoint boundary sets, and define

$$h(x) = \mathbb{P}_x(\tau_A < \tau_B).$$

Then

$$h|_A = 1, \quad h|_B = 0.$$

At every interior vertex x ,

$$h(x) = \frac{1}{\deg(x)} \sum_{y \sim x} h(y).$$

This is the first-step decomposition:

$$h(x) = \mathbb{E}_x[h(X_1)].$$

Hitting probabilities solve a Dirichlet problem.

Electrical side: voltages

Put unit resistors on the same graph.

Fix boundary voltages

$$v|_A = 1, \quad v|_B = 0.$$

At every interior vertex x , Kirchhoff's law gives

$$\sum_{y \sim x} (v(x) - v(y)) = 0.$$

Hence

$$v(x) = \frac{1}{\deg(x)} \sum_{y \sim x} v(y).$$

Therefore v satisfies the *same* harmonic equation and the *same* boundary conditions as h .

Voltages solve the same Dirichlet problem as hitting probabilities.

Uniqueness and the dictionary

Uniqueness principle. If two harmonic functions agree on the boundary, then they agree everywhere.

Indeed, if $u = v$ on the boundary, then $u - v$ is harmonic and vanishes on the boundary; by the maximum principle,

$$u - v \equiv 0.$$

Applying this to h and v , we obtain

$$h = v.$$

Thus the analogy becomes an identification:

hitting probability $\iff$ voltage

and more generally,

probabilistic questions $\iff$ electrical quantities.

From recurrence to resistance

Now consider the infinite lattice $\mathbb{Z}^d$, with unit resistors on all edges.

Let $R_{\text{eff}}(0 \leftrightarrow \infty)$ denote the effective resistance from the origin to infinity.

For simple random walk on $\mathbb{Z}^d$,

$$p_{\text{esc}} = \frac{1}{2d R_{\text{eff}}(0 \leftrightarrow \infty)}.$$

Hence

$$p_{\text{esc}} = 0 \iff R_{\text{eff}}(0 \leftrightarrow \infty) = \infty.$$

Since

$$\text{recurrent} \iff p_{\text{esc}} = 0,$$

we get

$$\text{SRW on } \mathbb{Z}^d \text{ is recurrent} \iff R_{\text{eff}}(0 \leftrightarrow \infty) = \infty.$$

So a return problem becomes a deterministic resistance estimate.

Rayleigh monotonicity

Shorting law. Identifying vertices can only decrease effective resistance:

$$R_{\text{eff}}(\text{shorted}) \leq R_{\text{eff}}(\text{original}).$$

Cutting law. Deleting edges can only increase effective resistance:

$$R_{\text{eff}}(\text{original}) \leq R_{\text{eff}}(\text{cut}).$$

For the proof of recurrence of $\mathbb{Z}^2$, we only need *shorting*.

Strategy: simplify the network while preserving a lower bound on the original resistance.

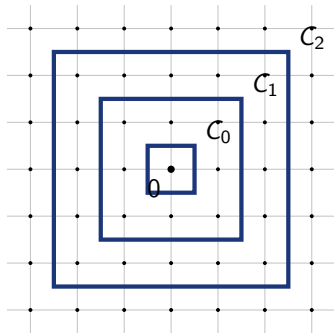
Application to $\mathbb{Z}^2$: square shells

Let

$$Q_n = [-n, n]^2 \cap \mathbb{Z}^2, \quad C_n = \partial Q_n.$$

We regard C_n as the n -th square shell around the origin.

The idea is to decompose the lattice into concentric square shells and then simplify the network shell by shell.



We will short together all vertices in the same shell C_n .

Application to $\mathbb{Z}^2$: shorting the shells

Short together all vertices in the same shell C_n .

By Rayleigh,

$$R_{\text{eff}}(\text{shorted network}) \leq R_{\text{eff}}(\mathbb{Z}^2).$$

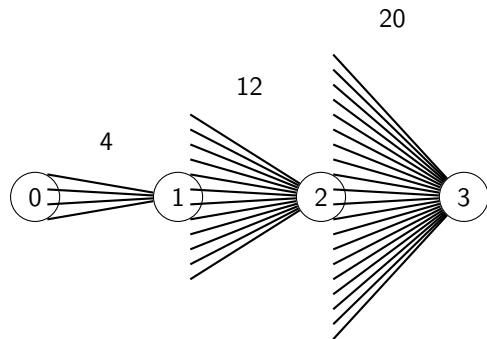
So it suffices to show that the shorted network already has infinite resistance.

Between C_n and C_{n+1} there are

$$8n + 4$$

parallel unit resistors.

For the first few layers, this gives



shorted network

Application to $\mathbb{Z}^2$: divergence of the resistance

Between C_n and C_{n+1} there are

$$8n + 4$$

parallel unit resistors.

So each layer is equivalent to a single resistor of value

$$\frac{1}{8n + 4}.$$

After shorting, the network becomes a series network, hence

$$R_{\text{eff}}(\text{shorted}) = \sum_{n=0}^{\infty} \frac{1}{8n + 4}.$$

Since this is a constant multiple of the harmonic series,

$$R_{\text{eff}}(\text{shorted}) = \infty.$$

Therefore

$$R_{\text{eff}}(\mathbb{Z}^2) \geq R_{\text{eff}}(\text{shorted}) = \infty,$$